TECHNICAL PORTFOLIO

Scientific Software and Research Engineering

Kai Woodger

Medical physics Computational modelling Reproducible software Evidence led validation

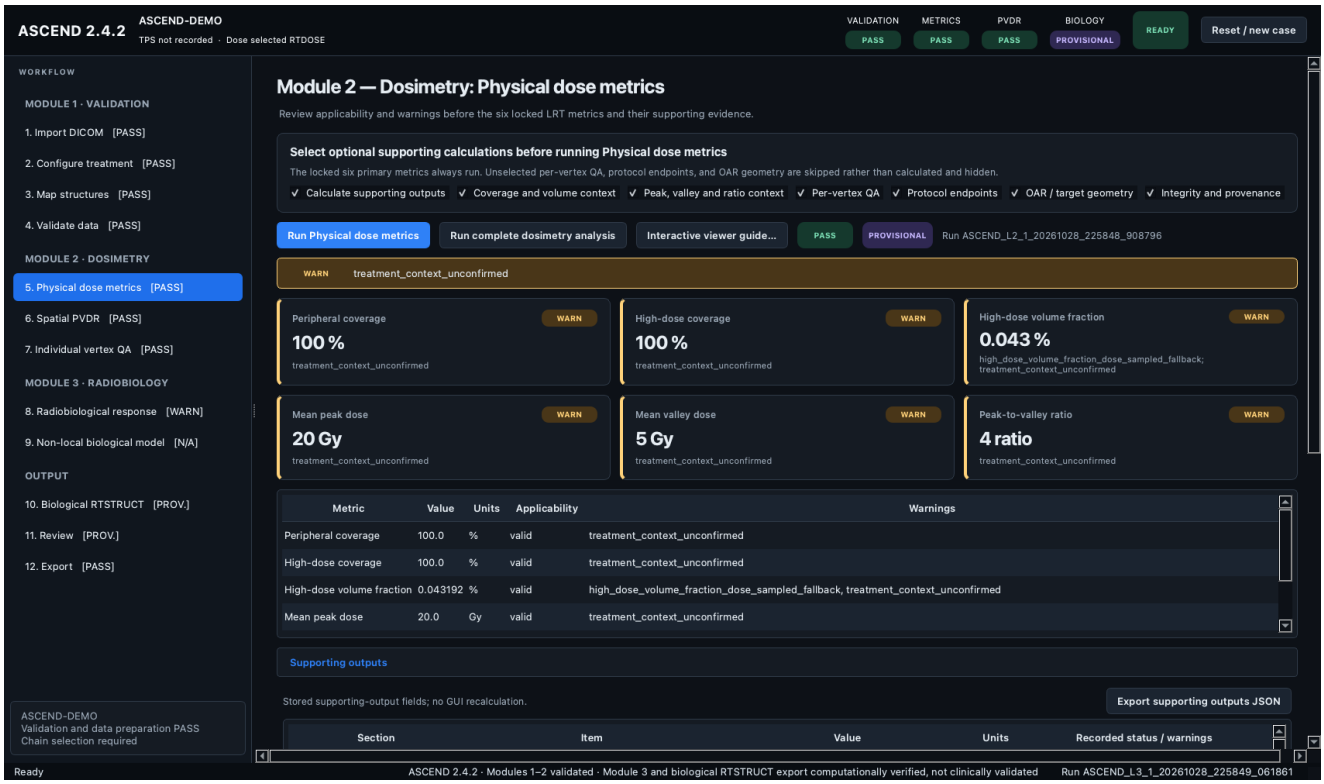


Figure 1 ASCEND 2.4.2 dosimetry workstation using a synthetic demonstration case

November 2026

Portfolio prepared from verified local repositories

PORTFOLIO SUMMARY

Portfolio Summary

Technical work verified against repository evidence

I develop research software that turns medical physics methods into auditable, reproducible tools. The work in this portfolio covers DICOM radiotherapy analysis, lattice radiotherapy metrics, computational digital twins, treatment planning automation, Monte Carlo beam modelling, biological audit systems, and machine learning quality assurance.

The strongest example is ASCEND, a cross platform Qt workstation for DICOM RT validation, lattice radiotherapy dosimetry, and research radiobiology. The current checkout passes 395 tests plus 40 subtests. Retained CI evidence spans six Linux, Windows, and macOS configurations, while frozen numerical and Eclipse comparisons quantify both agreement and unresolved failures.

Area	Evidence in this portfolio
Scientific software	ASCEND 2.4.2 desktop workstation and reproducible validation pipeline
Computational modelling	Deterministic multiscale radiotherapy digital twin with a watertight STL release
Treatment planning automation	C sharp ESAPI lattice sphere placement with geometry verification
Simulation and analysis	TOPAS VHEE beamline sweeps and transverse asymmetry analysis
Research platforms	BARAT biological audit framework and uncertainty aware contouring QA

Evidence standard Each case study separates software implementation from scientific or clinical validation. Research tools retain explicit warnings when commissioned measurements, commercial treatment planning systems, licensed anatomy data, or clinical outcomes are still required.

CASE STUDY 1

ASCEND Lattice Radiotherapy Workstation

Cross platform DICOM RT validation dosimetry and research radiobiology

Repository https://github.com/kaiwooodger/ASCEND

ASCEND is a Qt 6 desktop workstation that guides a case through three ordered modules. Module 1 validates the selected DICOM relationship chain and treatment context. Module 2 calculates locked physical lattice radiotherapy metrics, spatial PVDR, saddle evidence, and individual vertex QA. Module 3 contains computational research models whose outputs remain explicitly separate from clinically validated functions.

Current repository evidence	Value
Release	ASCEND 2.4.2
Current automated suite	395 tests plus 40 subtests passed on macOS ARM64 with Python 3.12
Cross platform matrix	229 tests plus 28 subtests passed on each of six Linux, Windows, and macOS configurations
Numerical reproducibility	70 baseline comparisons: 69 exact and one within tolerance; maximum absolute drift 8.88e-16
Eclipse benchmark	19 of 19 available dose comparisons passed; five of nine volume comparisons passed
Repository scale	166 application files, 44 test files, 84 documentation files, and four workflows

My contribution

- Defined the end to end scientific workflow and evidence boundaries for DICOM validation, dosimetry, spatial PVDR, radiobiology, and export.
- Integrated scientific services, GUI pages, state transitions, warnings, provenance, reporting, and synthetic fixtures into one workstation.
- Directed release engineering, cross platform testing, repository policy, documentation, and validation records.
- A prior Git audit attributed 99.68 percent of current code lines to Kai Woodger and 52 of 57 non merge commits, while recognising that repository attribution is not a measure of unaided manual typing.

ASCEND VISUAL EVIDENCE

ASCEND Interface and Scientific Method

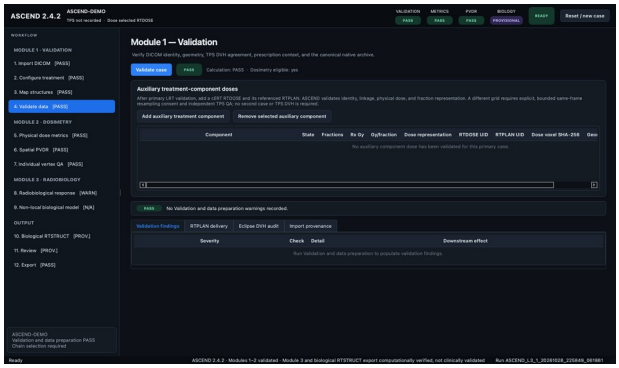


Figure 2 Module 1 validation interface

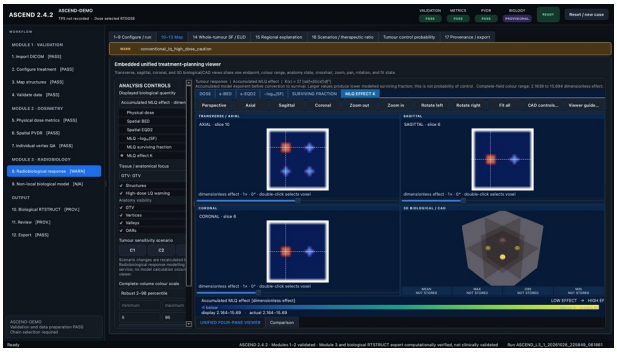


Figure 3 Module 3 research maps

The interface makes state and uncertainty visible. Disabled actions identify unmet prerequisites, warning panels name the required correction, and result exports retain version, commit, configuration hash, parameter identity, and source geometry evidence.

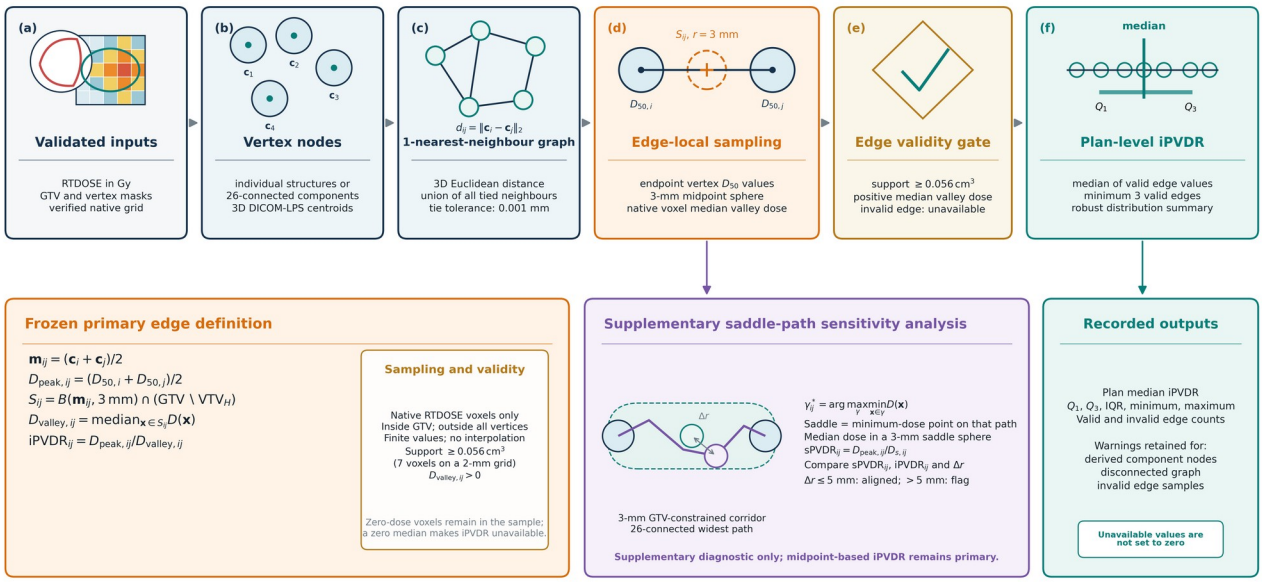


Figure 4 Publication figure for the edge local spatial iPVDR algorithm and supplementary saddle path analysis

ASCEND ENGINEERING

Verification Release and Research Boundaries

The 2.4.2 release adds ROI scoped auxiliary cERT coverage validation, preserves separate LRT and cERT fraction histories, and records incremental biological comparison evidence. The current checkout passed 395 tests and 40 subtests. A retained portability run passed 229 tests and 28 subtests on each of six Linux, Windows, and macOS configurations using Python 3.11 and 3.12.

Evidence class	What is established	Boundary retained
Automated suite	395 tests plus 40 subtests passed	Current local macOS and Python 3.12 execution
Cross platform	Six OS and Python configurations passed	Retained run used the earlier 229 test plus 28 subtest suite
Numerical accuracy	69 of 70 comparisons exact; maximum drift 8.88e-16	Frozen synthetic 14 metric reference
Eclipse dose	19 of 19 available comparisons passed	One stored case; eight D2 endpoints unavailable
Eclipse volume	Five of nine comparisons passed	Four failures remain explicit; no clinical validation claim

The Eclipse benchmark covered nine structures and 36 records. Twenty eight were valid comparisons, with 24 passes and four preserved structure volume failures. For the 19 available dose comparisons, the median absolute difference was 0.00643 Gy and the maximum was 0.22245 Gy under the provisional max 0.2 Gy or 2 percent rule. This is quantitative external TPS evidence, not complete formal or clinical validation.

Portfolio value ASCEND demonstrates that I can combine medical physics, DICOM data structures, numerical methods, user interface design, software architecture, testing, documentation, and evidence based release management in one working system.

- DICOM RT identity and geometry handling
- Lattice radiotherapy plan quality and spatial dose analysis
- Qt desktop application design and workflow state management
- Scientific regression testing and synthetic fixtures
- Publication figures reports and thesis documentation
- GitHub Actions release contracts and cross platform portability

CASE STUDY 2

Dynamic Multiscale Radiotherapy Digital Twin

Deterministic engineering baseline for anatomy motion circulation radiation and biology

This project implements a reproducible whole body computational framework with explicit DICOM LPS coordinates, deformation, circulation, blood packet transport, lymph transport, radiation energy accumulation, oxygen fields, biological response interfaces, uncertainty propagation, checkpoints, and provenance manifests.



Figure 5 Watertight synthetic whole body mesh

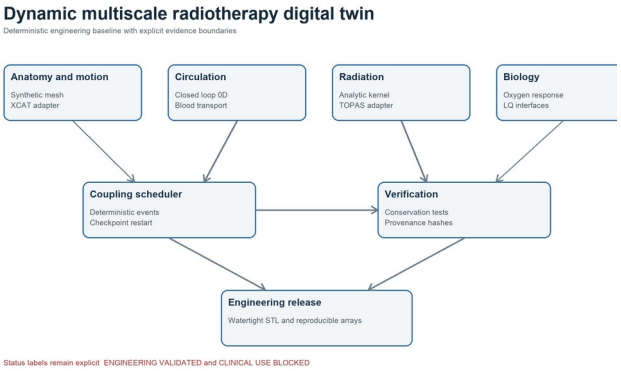


Figure 6 Implemented system architecture

Engineering release evidence	Result
Mesh	39,664 triangles
Topology	Watertight with zero boundary nonmanifold or degenerate elements
Automated tests	8 passed in the immutable v0.1.0 release
Clean rerun	Simulation arrays summaries and STL were byte identical
Status	ENGINEERING VALIDATED and CLINICAL USE BLOCKED

DIGITAL TWIN VALIDATION

External Reference Integration and Evidence Gates

The project treats external references as scoped validation layers. HEDOS tests blood transport and blood dose behaviour. XCAT tests anatomy and motion interfaces when a licensed executable and reference subject are available. TOPAS provides a boundary for commissioned radiation transport. Passing one layer never grants claims that belong to another.

Reference	Verified result	Remaining boundary
HEDOS	Twenty seeds and 100,000 particles per seed reproduced field and DVH based modes	Does not establish human physiology
XCAT	Adapter synthetic transforms and 24 of 24 authorised package intake criteria passed	Execution blocked by absent provider licence and Windows x86 64 requirement
TOPAS	Locked runtime smoke case and adapter evidence	Independent measurement and site commissioning still required
Coupling	Stationary moving dose conservation restart and refinement checks	Synthetic interface verification only

My contribution

- Converted a broad radiotherapy digital twin specification into versioned modules data contracts benchmarks and acceptance criteria.
- Built deterministic execution and reproducibility controls around geometry physiology transport radiation and biological interfaces.
- Preserved licensing computational platform and clinical evidence constraints rather than overstating readiness.
- Advanced the later repository to 65 passing tests during authorised XCAT intake while retaining the clinical block.

CASE STUDY 3

ESAPI Lattice Sphere Placement

C sharp treatment planning automation with independent geometry evidence

Repository https://github.com/kaiwooodger/ESAPI-Script

The script implements Gaudreault style three dimensional sphere lattice placement for an Eclipse structure set. It creates individual control structures and a union while preserving the original target, enforcing target contraction, image bounds, spacing, containment, output naming, and cleanup rules.

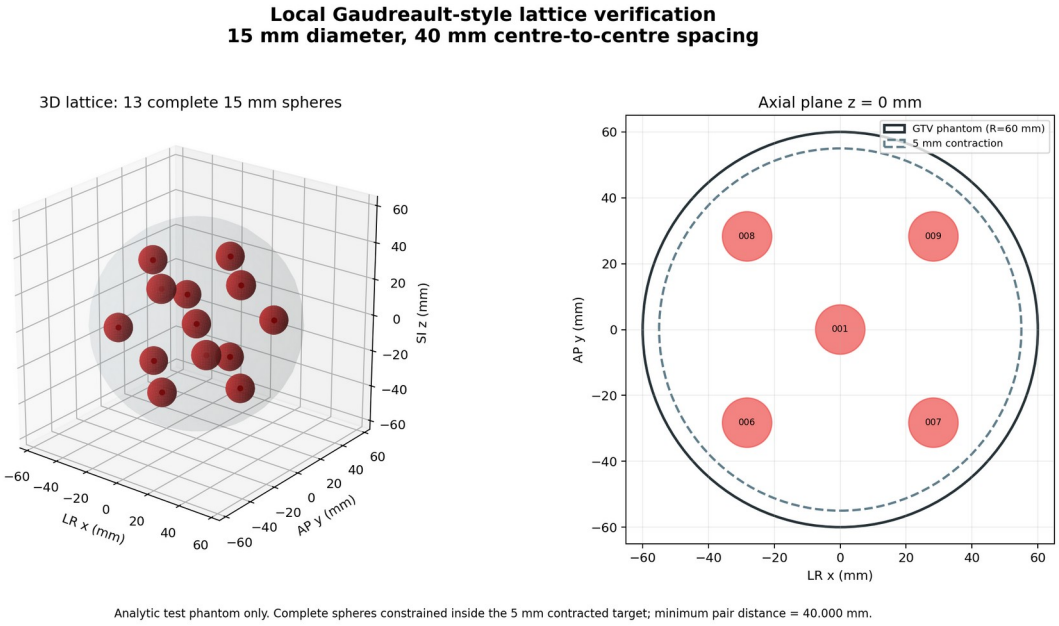


Figure 7 Local phantom verification for 15 mm spheres with 40 mm centre to centre spacing

- Shared geometry was compiled directly from the deployed C sharp source.
- A worked phantom produced 13 complete spheres with a 40.000000 mm minimum pair distance.
- The original GTV remains unchanged and existing output structure identifiers cause a pre modification failure.
- Actual Eclipse execution and compilation against installed Varian assemblies remain installation specific validation steps.

ESAPI VERIFICATION

Parameter Sweep and Robustness Evidence

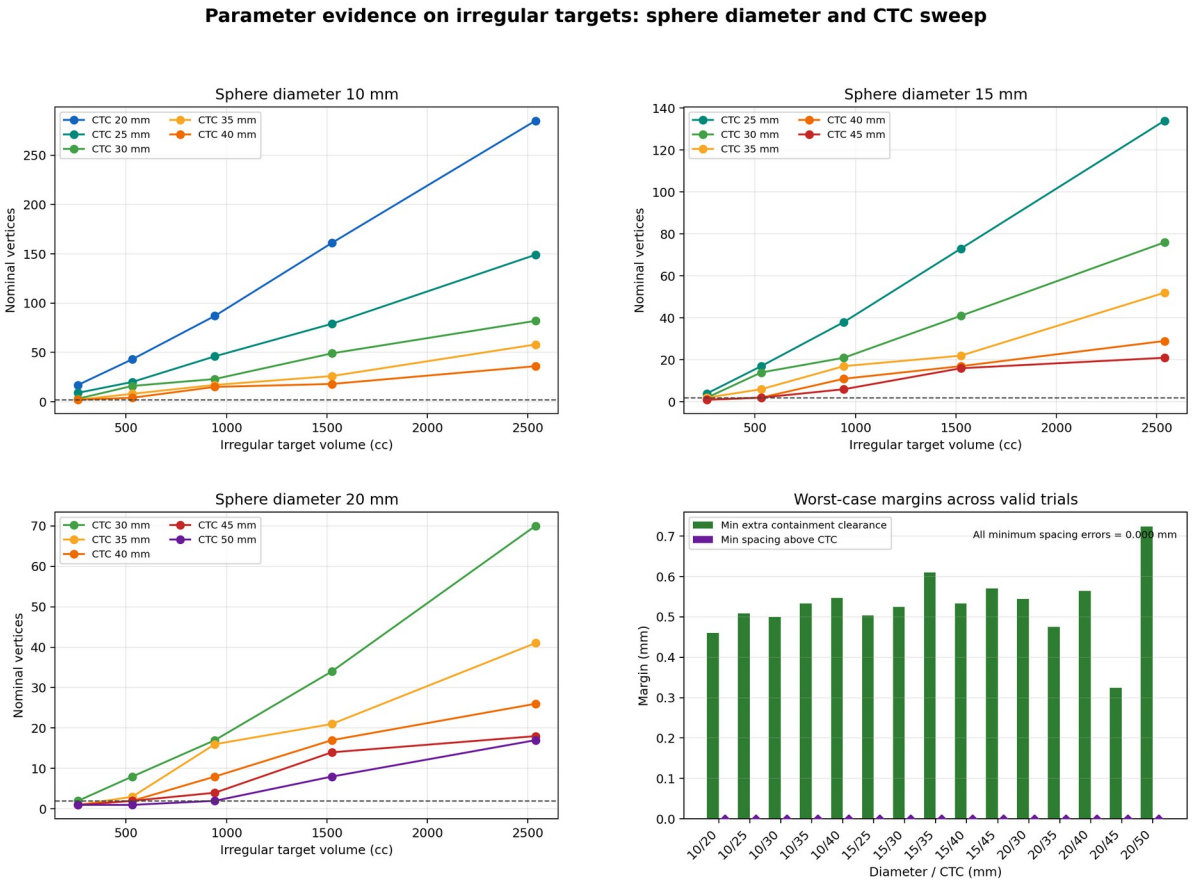


Figure 8 Sphere diameter and centre to centre spacing evidence across irregular target sizes

Verification measure	Recorded result
Total trials	375
Valid lattices	336
Correct no lattice outcomes	39
Geometric failures	0
Worst spacing error	0.000000 mm
Minimum containment clearance	0.324 mm
Maximum tested lattice	292 vertices
Core regression suite	19 passed

ADDITIONAL PROJECTS

Biological Audit Contouring QA and VHEE Simulation

Archived project evidence and reproducible analysis outputs



Figure 9 BARAT verification evidence

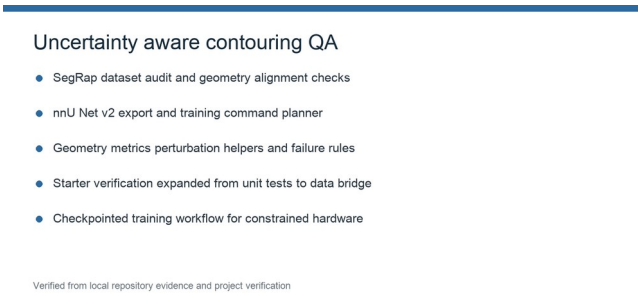


Figure 10 Contouring QA verification evidence

BARAT Biological Audit and Risk Analysis Tool

BARAT was structured as a phased CLI and dashboard research platform for biological plan audit. The Phase 8 interface exposed data readiness, model trust, calibration lock, plan comparison, report exports, and an explicit clinical use gate. Recorded verification reached 63 to 64 passing tests, and the repository was prepared for GitHub with workflow checks, data handling rules, citation metadata, and release guidance.

Uncertainty Aware Contouring QA

The contouring QA project tests the claim that automatic contours should be judged by stability and plausibility as well as average overlap. The scaffold includes geometry metrics, perturbation helpers, failure rules, SegRap data auditing, an nnU Net v2 export bridge, training command planning, notebooks, and checkpointed workflows for constrained compute environments.

SIMULATION RECORD

TOPAS VHEE Beamline Modelling

The VHEE project built TOPAS scripts and Python analysis for a four quadrupole asymmetric beamline at 100 200 and 250 MeV. Development included Q4 sweeps, source phase space sensitivity, integrated depth metrics, Gaussian transverse width analysis, reproducible manifests, and plots that separate depth dose from lateral asymmetry.

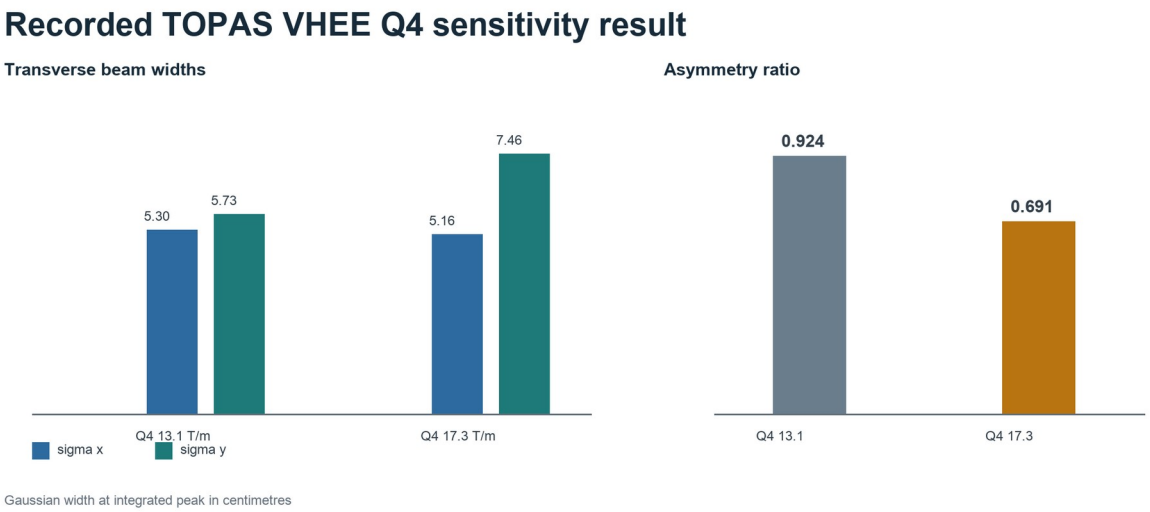


Figure 11 Recorded 250 MeV Q4 sensitivity result using the 4.0 mm source sigma baseline

The recorded analysis showed that Q4 equal to 17.3 T per m increased asymmetry compared with 13.1 T per m. At the integrated peak the Gaussian width ratio changed from 0.924 to 0.691. The work also established that a one dimensional percentage depth dose curve cannot demonstrate transverse asymmetry by itself.

Q4 T per m	Integrated peak cm	Sigma x cm	Sigma y cm	Ratio
13.1	23.9109	5.2958	5.7307	0.924
17.3	16.7822	5.1558	7.4577	0.691

PROFESSIONAL SUMMARY

Technical Capability Matrix

Evidence that can be used directly in a resume or application

Capability	Demonstrated evidence
Medical physics software	DICOM RT workflow validation lattice radiotherapy dosimetry spatial PVDR radiobiology and reporting
Scientific computing	Python NumPy SciPy PySide6 PyVista pydicom R and reproducible numerical workflows
Treatment planning	Eclipse ESAPI C sharp automation DICOM RTSTRUCT logic and synthetic phantom verification
Computational modelling	Anatomy motion circulation blood and lymph transport radiation oxygen and biological response interfaces
Monte Carlo simulation	TOPAS beamline construction parameter sweeps manifest based execution and dose grid analysis
Machine learning QA	SegRap data audit nnU Net v2 bridge uncertainty metrics failure rules and checkpointed training
Software engineering	Modular architecture tests CI release records package checks provenance schemas and documentation
Research communication	Publication figures thesis material validation appendices abstracts and evidence led technical reporting

Resume ready project bullets

- Developed ASCEND 2.4.2, a cross platform Qt workstation whose current suite passes 395 tests plus 40 subtests; retained CI verifies six Linux, Windows, and macOS configurations, and 19 of 19 available Eclipse dose comparisons pass the provisional agreement rule.
- Designed a deterministic multiscale radiotherapy digital twin integrating synthetic anatomy, motion, closed loop circulation, transport, radiation, oxygen response, checkpoint restart, and evidence gated validation.
- Implemented and verified a C sharp ESAPI lattice sphere placement engine across 375 irregular target trials with zero geometry failures among 336 valid lattices.
- Built a TOPAS VHEE beamline and analysis pipeline for energy and quadrupole sweeps, quantifying transverse asymmetry and separating lateral beam behaviour from depth dose metrics.
- Created BARAT and uncertainty aware contouring QA research platforms with reproducible CLI workflows, model trust states, data readiness checks, nnU Net integration, and explicit clinical use gates.

Development approach I use AI assisted engineering tools for implementation, debugging, validation, documentation, and release work. I retain responsibility for scientific scope, source selection, acceptance criteria, interpretation, risk boundaries, and final technical decisions.